

Abigail Lin

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Education

Doctor of Philosophy in Computer Science | University of Florida

August 2026 - Present

GPA: N/A

Bachelor of Science in Computer Science | Math Minor | University of Florida

August 2022 - May 2026

Research Experiences

University of Florida, Undergraduate Research

August 2025 - May 2026 | Xie Lab

- Fine-tuned the protein language model ProGen for generation of Type-III bacterial PKS to explore the protein space for potential novel natural products.

Purdue University, RCAC Anvil REU

May 2025 - August 2025 | Rosen Center For Advanced Computing

- Designed and implemented a bioinformatics workflow template generator (Bash, Slurm, Python, Urwid) providing a simpler, more accessible alternative to Nextflow and Snakemake.
- Implemented variant calling and genome assembly pipelines.
- Used Apptainer and Docker for container creation and deployment to efficiently organize and manage software dependencies.

University of Florida, AI Scholars Program

August 2024 - May 2025 | Voiniciuc Lab | Xie Lab

- Generated 30 CSLA enzyme variants using a Maximum Entropy (MaxEnt) probabilistic AI model on UF's HiPerGator supercomputer.
- Automated experimental assays to evaluate the efficiency of generated enzyme variants in synthesizing β -mannans.
- Benchmarked ProteinGym mutation scores using generative protein language models, including ESM2 and SaProt.

University of Central Florida, Applied Computational Mathematics NSF REU

May 2024 - July 2024 | Dr. Carlos Borges

- Worked for 10 weeks as an NSF-funded summer undergraduate researcher.
- Developed a MATLAB solver for the forward scattering problem of objects with thin coatings, comparing impedance and transmission approximations and achieving a final error below 10^{-2} .

University of Florida, Research Assistant

January 2024 - May 2026 | Voiniciuc Lab

- Developed Python programs to automate synthetic biology workflows using four Opentrons OT-2 lab robots, reducing pipetting error and increasing experimental efficiency.
- Wrote Python scripts to automate batch editing of GenBank files in Benchling, leveraging Biopython and Pandas for data parsing and processing.

- Conducting research on generative AI for plant enzyme engineering, applying protein language models to design enhanced β -mannanase (MAN) enzymes.

Publications

Qiande, M., **Lin, A.**, Larson, L., & Voiniciuc, C. (2025). BOTany Methods: Accessible Automation for Plant Synthetic Biology. *Plant Physiology*, 2025-08.

Awards

UF HWCoe Dean's Research Award, University of Florida (2026 - 2027)

Standing Innovation Award, University of Florida Innovate (2025) - Recognized for contribution to the invention disclosure "Evolved Cellulose Synthase-Like Enzymes for Polysaccharide Production."

Presentations

Summer Undergraduate Research Symposium

July 2025 | Purdue University

- "Empowering Research: Easy-to-Use Bioinformatics Workflow Templates," *Undergraduate Research Symposium*, Purdue University, West Lafayette, Indiana, July 2025.

2025 UF Spring Undergraduate Research Symposium

April 2025 | University of Florida

- "Using the Maximum Entropy Model to Enhance CSLA Enzyme β -Mannan Polysaccharide Synthesis," *2025 UF Spring Undergraduate Research Symposium*, University of Florida, Gainesville, Florida, April 2025.

Summer Poster Showcase

July 2024 | University of Central Florida

- "Forward Scattering for Thin Coated Domains: Numerical Solution Using Generalized Impedance Boundary Conditions," *Summer Poster Showcase*, University of Central Florida, Orlando, Florida, July 2024

Involvement

Phi Sigma Rho

January 2024 - May 2026

- Participated in various sorority events, including volunteering, fostering a supportive community for women in STEM fields.

Women in Computer Science and Engineering (WiCSE)

September 2023 - December 2023

- Mentored a computer science major underclassman, provided guidance and technical instruction as well as, fostering their academic and professional growth.